$$\sigma_z|r\rangle = \frac{1}{\sqrt{2}}|u\rangle - \frac{1}{\sqrt{2}}|d\rangle. \qquad (3.21)$$

OK, here is our trap. Despite what you might think, the state-vector on the right-hand side of Eq. 3.21 is definitely *not* the state that would result from a measurement of σ_z. That measurement result would be either +1, leaving the system in state $|u\rangle$, or −1, leaving it in state $|d\rangle$. Neither of these results would leave the system state-vector in the superposition represented by Eq. 3.21.

But surely that state-vector must have *something* to do with the measurement result? In fact, it does. We'll find part of the answer in Lecture 4, where we'll see how the new state-vector allows us to calculate the probabilities of each possible outcome of the measurement. However, the result of a measurement cannot be properly described without taking the apparatus into account as part of the system. What actually does happen during a measurement is the subject of Section 7.8.

3.6 3-VECTOR OPERATORS REVISITED

Now, let's revisit the idea of a 3-vector operator. I have called σ_x, σ_y, and σ_z, the components of spin along the three axes, implying that they are the components of some kind of 3-vector. This is a good time to return to the two notions of vectors that come up all the time in physics. First, there is your garden-variety vector in ordinary three-dimensional space, which we've decided to call a 3-vector. As we've seen, a 3-vector has components along the three directions of space.

The other completely distinct meaning of the term *vector* is the state-vector of a system. Thus, $|u\rangle$ and $|d\rangle$, $|r\rangle$ and $|l\rangle$, and $|i\rangle$ and $|o\rangle$ are state-vectors in a two-dimensional space of spin states. What about σ_x, σ_y, and σ_z? Are they vectors, and if so, what kind?

Clearly, they are not state-vectors; they are operators (written as matrices) that correspond to the three measurable components of spin. In fact, these 3-vector operators represent a new type of vector. They are different both from